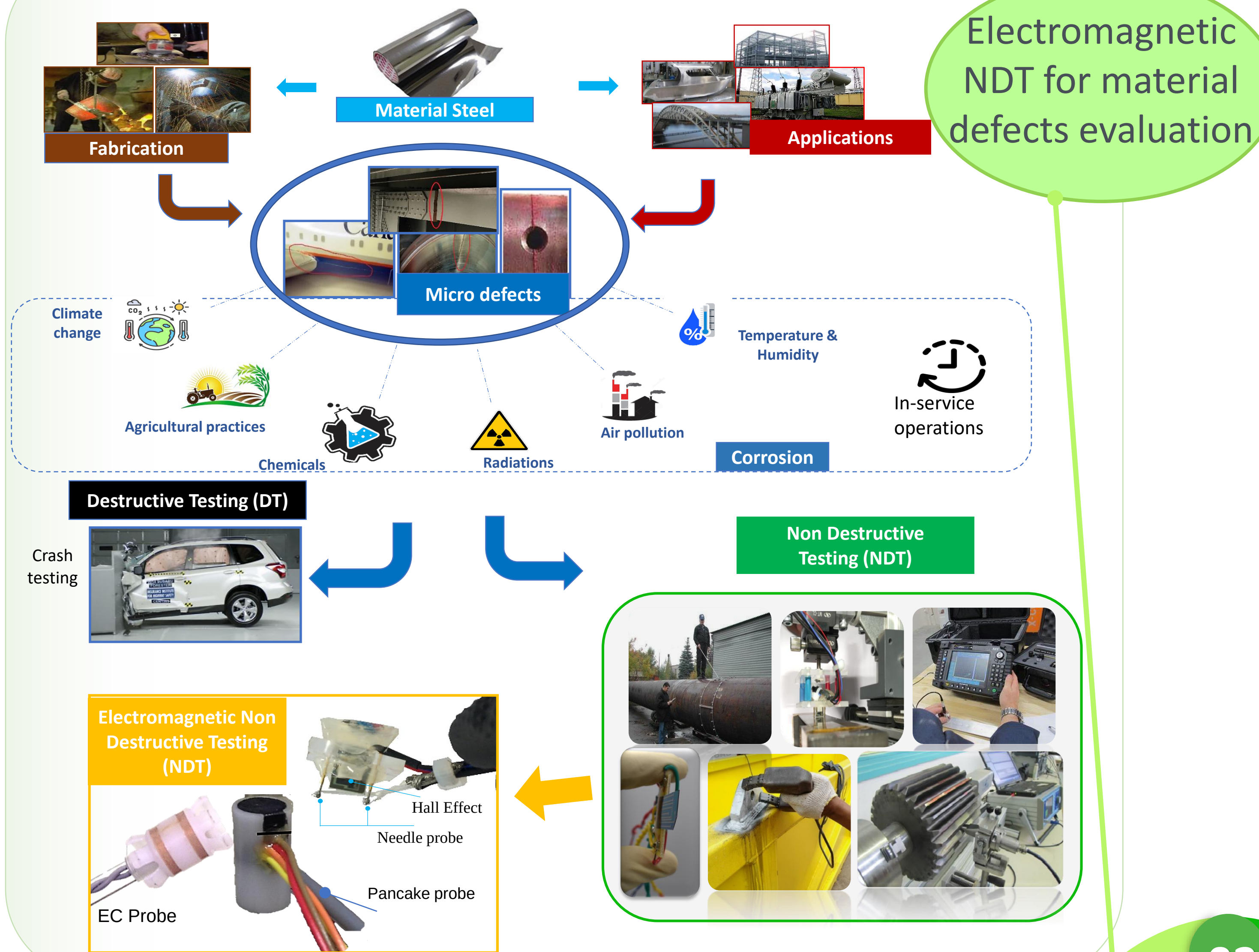


Executive Summary

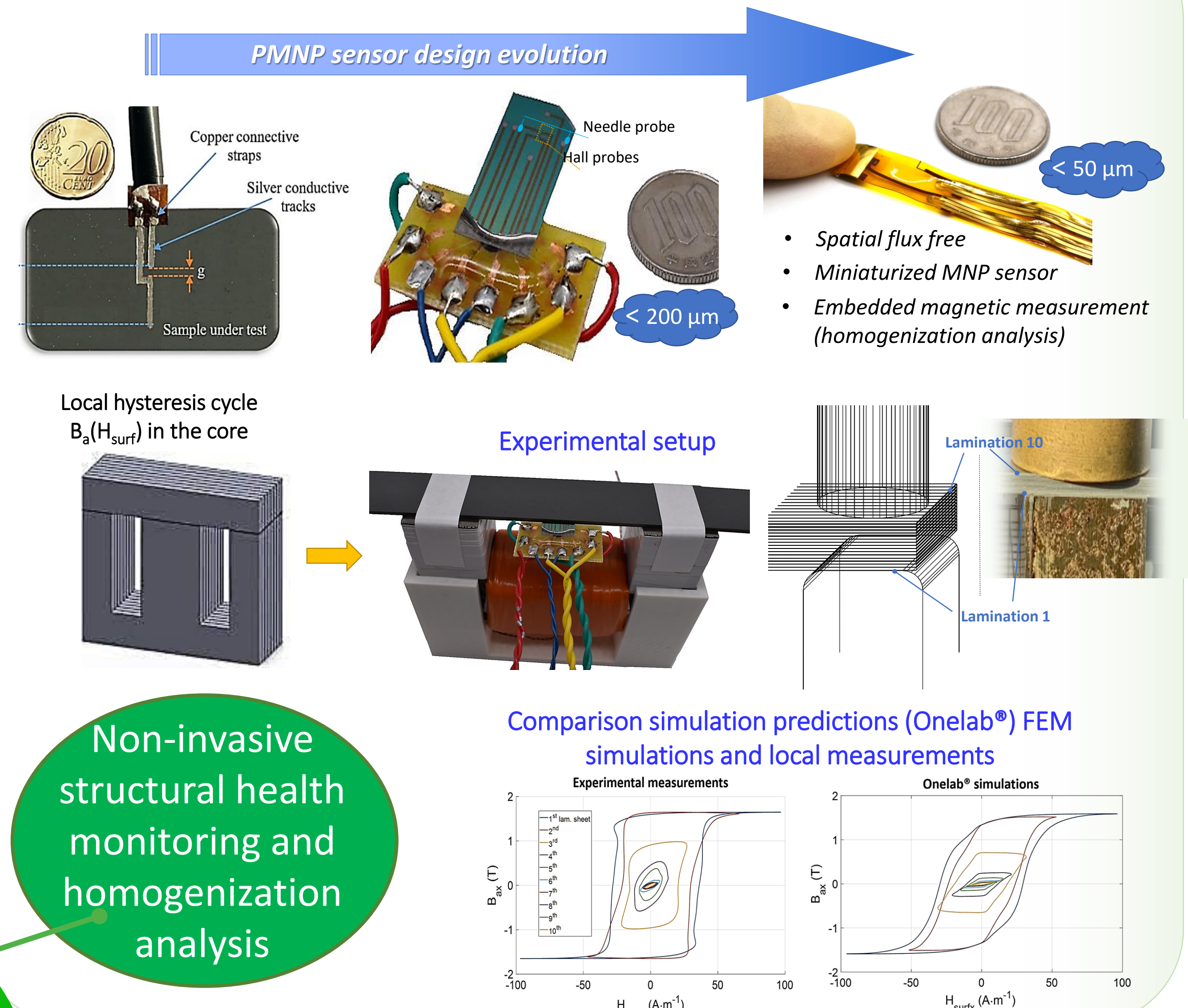
Every engineering system experience significant performance depreciation in its service time till failure of essential parts. This drop in performance informs of the constant evolution of material microstructural parameters such as grains size, orientation, stress and strain fields, and eventually shapes the goals of the current emerging electromagnetic NDT methods towards material integrity challenges. The field of NDT is evolving. This evolution is related to the complexity of the components to be controlled and to the improvement of the instrumentation and the computing techniques. Eddy current testing (ECT), magnetic Needle Probe(MNP), Magnetic Incremental Permeability (MIP) and magnetic Barkhausen noise (MBN) are some examples of electromagnetic NDT methods with numerous applications in piping, aeronautics, electrical machinery, etc. Exploiting the magnetic behaviour of material steels is crucial for the accurate simulation of electromagnetic devices and eventually, an accurate prediction of their performances.

Our motivation

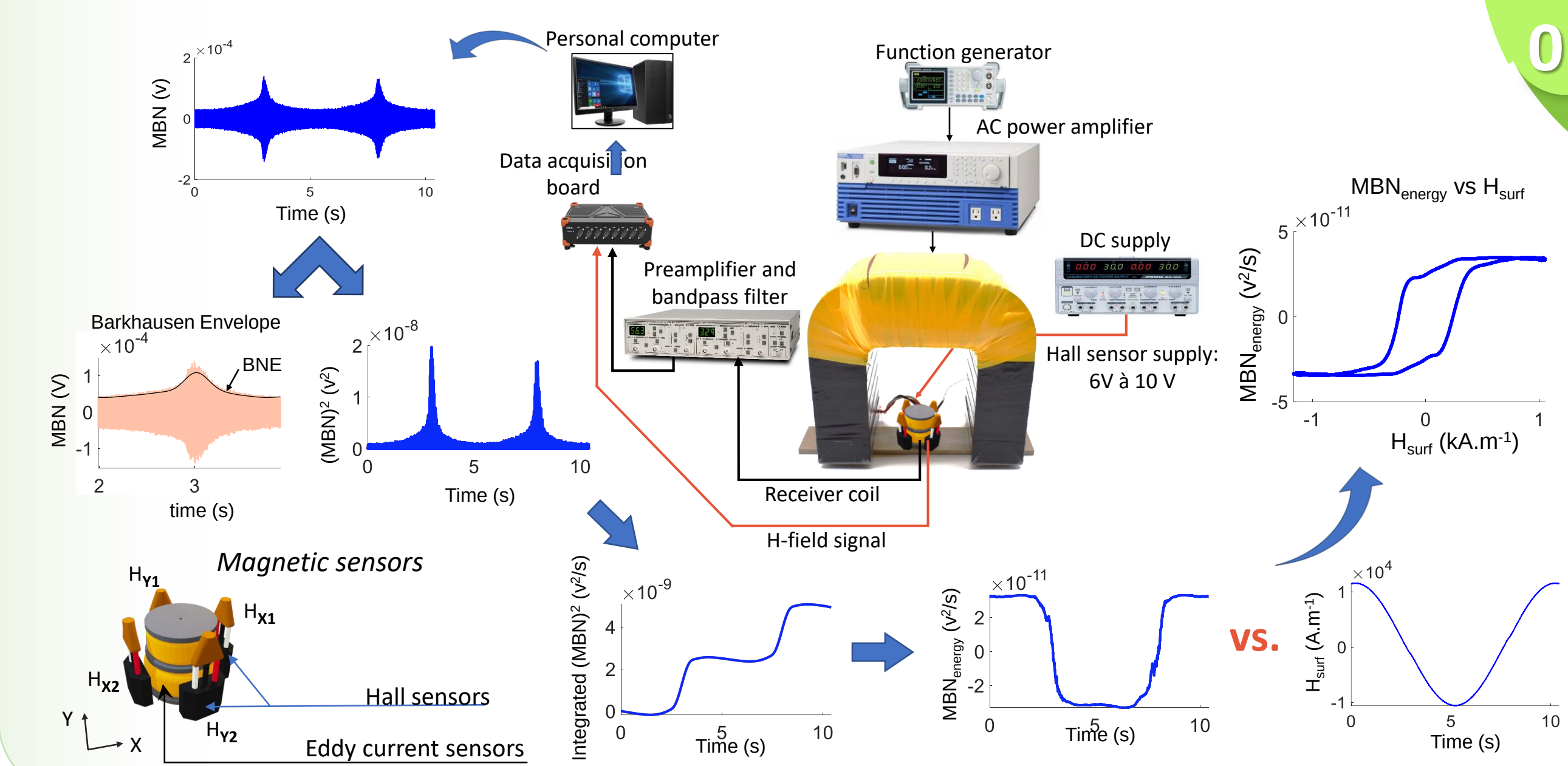


Printed Magnetic Needle Probe (PMNP)

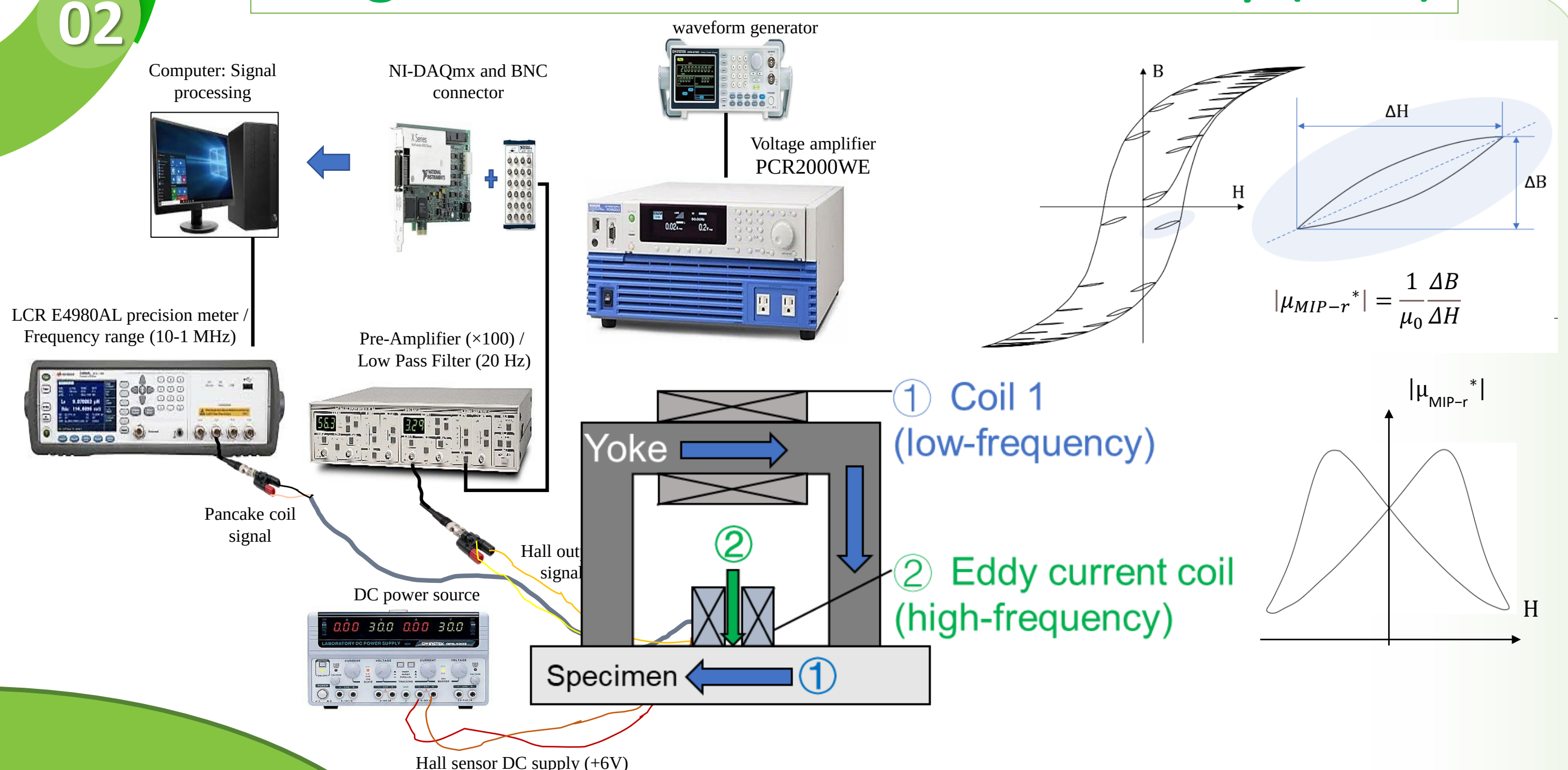
➤ **Objectif:** Internal characterization of a laminated magnetic core



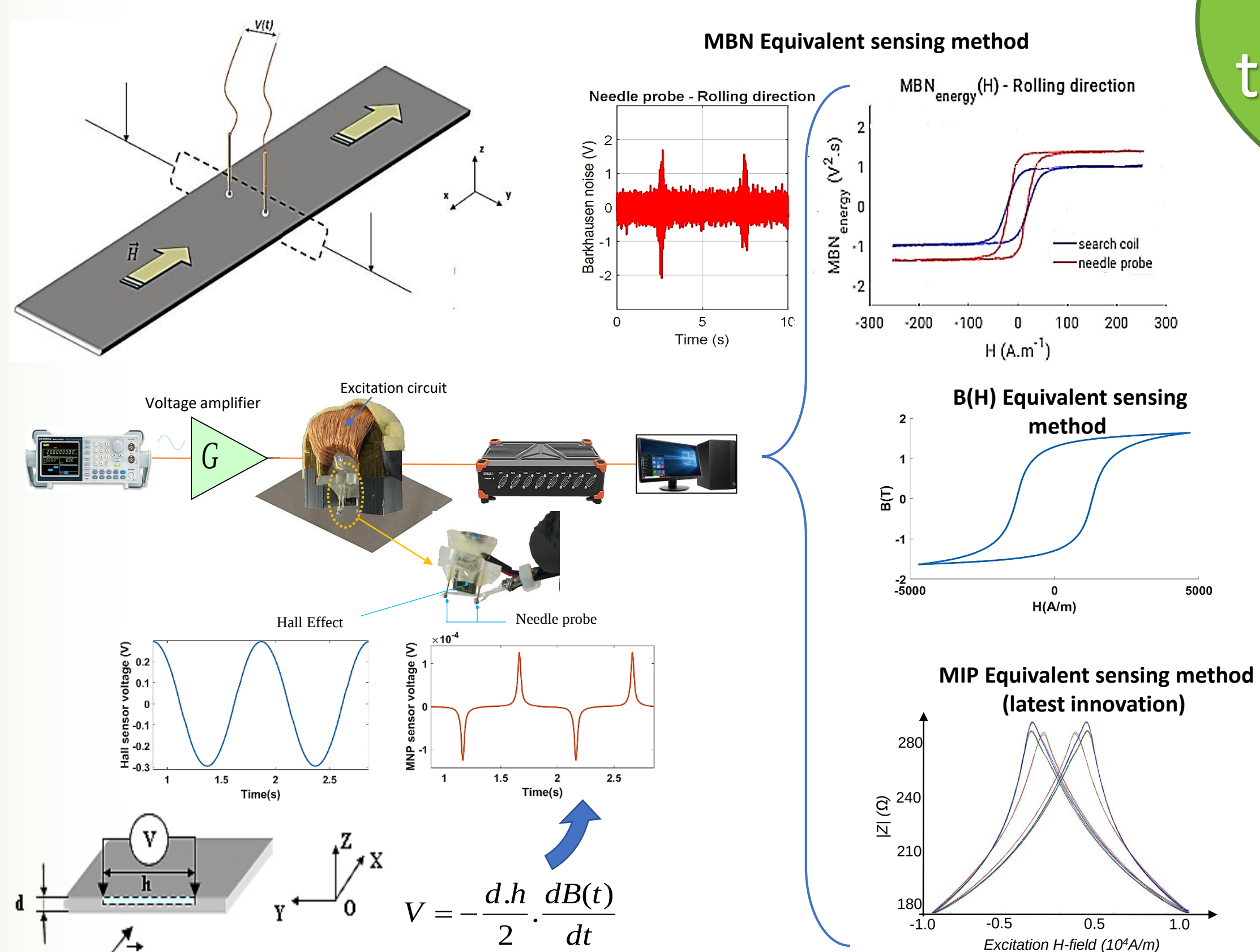
Magnetic Barkhausen Noise (MBN)



Magnetic Incremental Permeability (MIP)

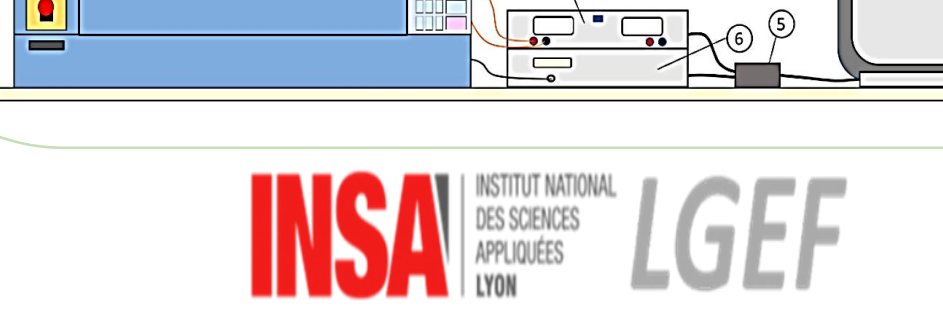
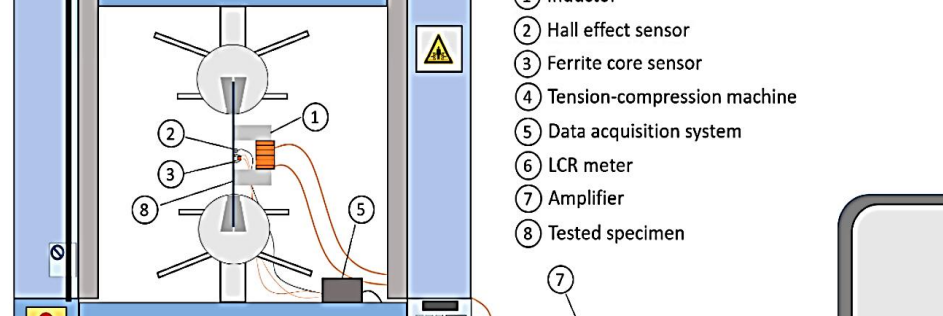
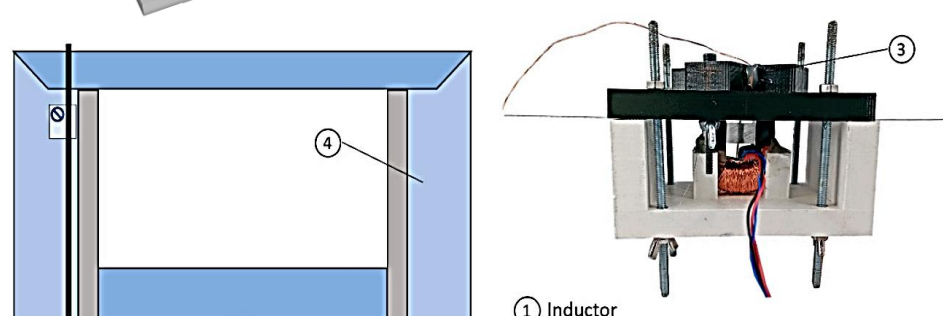
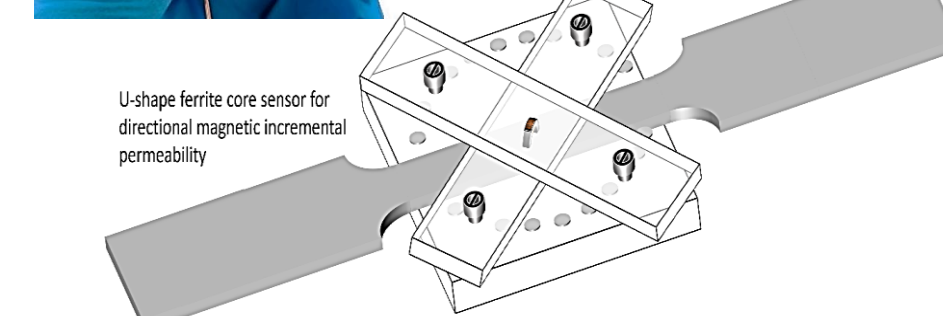
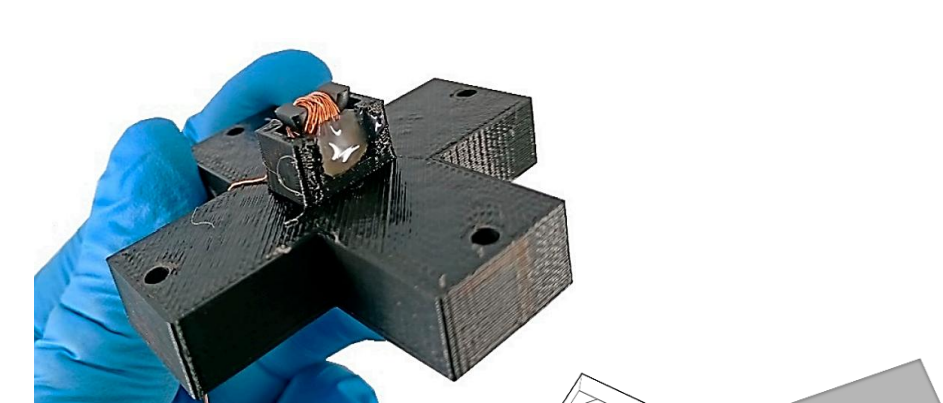


Magnetic Needle Probe (MNP)



Design of electromagnetic instrumentation methods towards integrity evaluation of materials and systems

Characterization of Tensile Stress-Dependent Directional Magnetic Incremental Permeability in Iron-Cobalt



Smart NDT Applications

